Q $y = \arctan(x)$ $\quad \frac{dy}{dx}?$

A $\tan y = x$

$\frac{d}{dx} \downarrow$

$\sec^2 y \cdot \frac{dy}{dx} = 1$

$\frac{dy}{dx} = \frac{1}{\sec^2 y}$ ← Right side only has y.

Side $\sin^2 y + \cos^2 y = 1$

$\frac{\sin^2 y}{\cos^2 y} + \frac{\cos^2 y}{\cos^2 y} = \frac{1}{\cos^2 y}$

$\tan^2 y + 1 = \sec^2 y$

$\frac{dy}{dx} = \frac{1}{\tan^2 y + 1}$

$\frac{dy}{dx} = \frac{1}{x^2 + 1}$